

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology & Geophysics

2. Subject Name: Petrology L-T-P: 3-0-0 Credits: 3

3. Pre-requisites: Introduction to Earth Science

4. Syllabus and reference books:

Syllabus: Igneous: Nature of silicate melts and crystallization processes; Classification of igneous rocks; Interpretation of elementary liquidus phase diagrams; Magmatic differentiation and its importance in the genesis of igneous rocks; Plate tectonics and magmatism.

Sedimentary: Classification of sedimentary rocks; weathering and erosion, sedimentary textures and structures and relationship to dynamics of flow and deposition; Physics of sedimentary basin formation.

Metamorphic: Definition, metamorphic facies classification and baric types; Metamorphic variables; Metamorphism of mafic and pelitic rocks; Metamorphism in relation to plate tectonics.

Reference Books:

- 1) Principles of Igneous and Metamorphic Petrology – A. Philpotts and J. J. Ague
- 2) Introduction to Sedimentology – S. M. Sengupta

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Igneous Petrology	14
2.	Sedimentary Petrology	14
3.	Metamorphic Petrology	14
Total number of hours		42